

HW-2

Packages

Question 1

```
install.packages("aRtsy")
```

`install.packages()` downloads and installs a package on your computer. `library()` loads an installed package into the current R session.

Question 2

```
library(aRtsy)
```

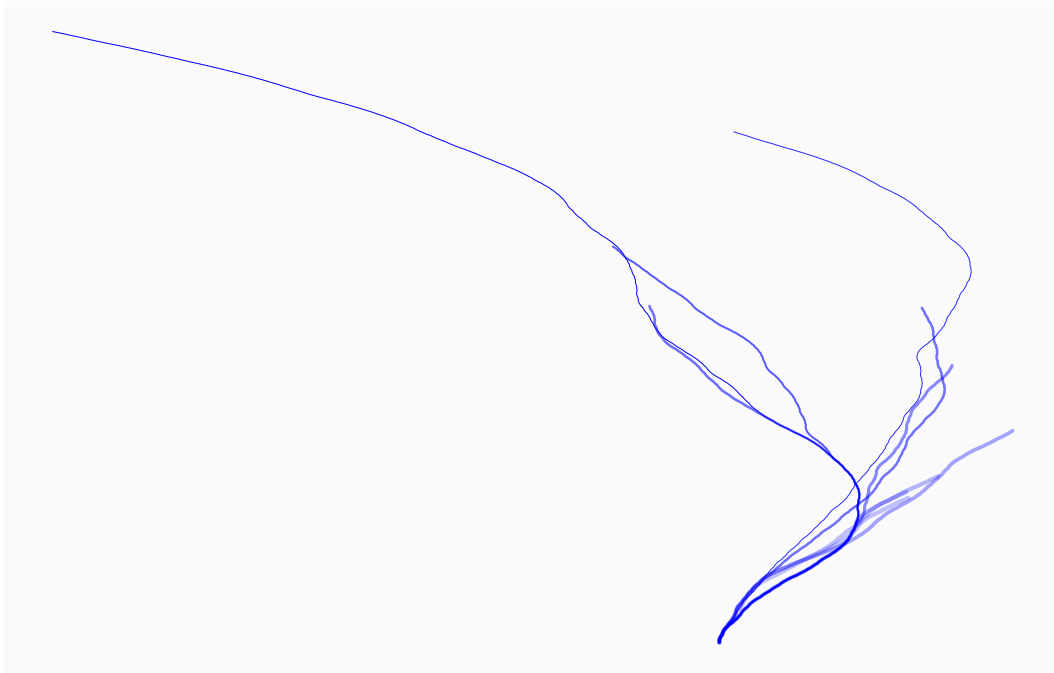
The Collatz conjecture starts with any positive integer n , if n is even, divide it by two; otherwise, multiply it by 3 and add 1. Repeats this process with the new n value. Eventually, n will reach to 1.

Let $f(n)$ be the next number obtained from n :
$$f(n) = \begin{cases} n/2, & \text{if } n/2 = 0, \\ 3n + 1, & \text{if } n/2 = 1. \end{cases}$$

Question 3

3.a

```
set.seed(7)
canvas_collatz(
  colors = c("blue"),
  n = 10
)
```



`canvas_collatz()` creates a plot of Collatz sequences using blue lines. `n = 10` tells it to choose 10 random starting integers.

3.b

`canvas_collatz()` chooses 10 random starting integers, so it generates a different plot every time we run the code. This is troublesome in `.qmd` file because rendering again could change the plot, making the outcome hard to reproduce and compare. The `set.seed()` sets the same seed before generating random numbers gives the same outcome each time.

Question 4

```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```



In question 3, `n = 10` tells `canvas_collatz()` to choose 10 random starting numbers. In question 4, we use a vector `n = 1:20`, which tells it to use the number 1 to 20 as starting numbers. This argument makes `canvas_collatz()` to generate the same plot each time we render. Since the starting numbers are fixed, including `set.seed()` is unnecessary: the seed would not change this plot.